Evaluation Methods in NLP

Evaluation paradigms in NLP

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Parameswari Krishnamurthy

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

Evaluation

Assessing the performance and effectiveness of NLP models, systems, or algorithms.

Focused on understanding and generating human language.

Purpose of Evaluation:

- Ensure models meet task-specific goals.
- Identify strengths and weaknesses for improvement.

Evaluation Techniques:

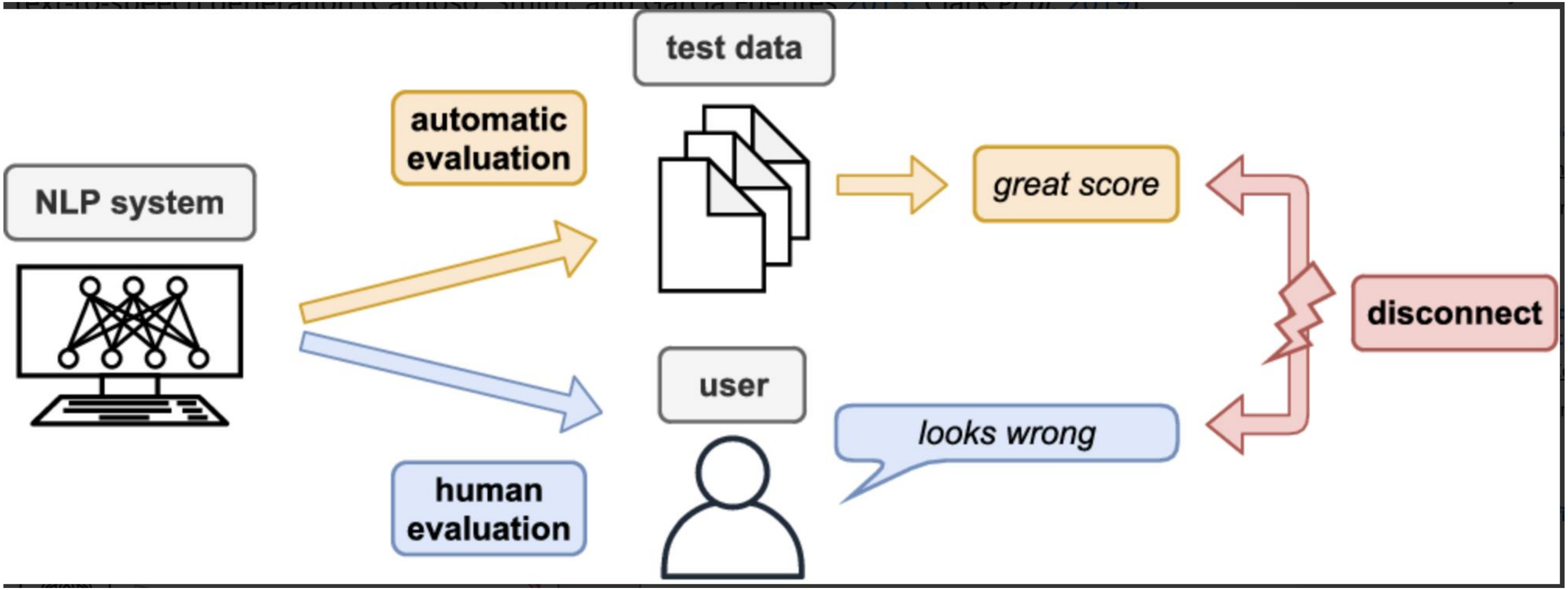
- Vary based on the specific task (e.g., translation, sentiment analysis, etc.).
- Use a combination of **metrics** (e.g., Precision, Recall) and **methods** (e.g., human reviews).

Why do we evaluate?

- Baking a cake
- **Goal**: A moist and flavorful chocolate cake.
- **Metrics**: Taste, texture, and appearance.
- **Evaluation**:
  - Taste test (subjective, human evaluation).
  - Measure texture (objective: sponge density).
- **Results**:
  - Cake is visually appealing but lacks flavor → Improve recipe.

Aspect	Cooking Evaluation	NLP Evaluation
Goal	Assessing the quality of the dish (e.g., taste, texture).	Measuring model performance on tasks (e.g., translation, summarization).
Evaluation Metrics	Taste, texture, appearance (subjective and qualitative).	Precision, Recall, F1-Score, BLEU, ROUGE (objective and quantitative).
Input	Ingredients and recipe.	Dataset (training, validation, and test splits).
Output	A cooked dish (e.g., a cake).	Model predictions (e.g., translations, classifications).
Evaluation Method	Human feedback (taste test) and visual inspection.	Automated comparison of predictions with ground truth data.
Example	A cake is moist but lacks flavor, requiring recipe adjustment.	A machine translation model produces errors in grammar → Adjust model architecture or data.

Does automatic evaluation solve the issue?



Foundation for Matrics in classification

True Positives (TP) :

- Interpretation: Model predicted positive and it’s true.

False Positives (FP):

- Interpretaion: Model predicted positive and it’s false.

True Negatives (TN):

- Interpretation: Model predicted negative and it’s true.

False Negatives (FN):

- Interpretation: Model predicted negative and it’s false.

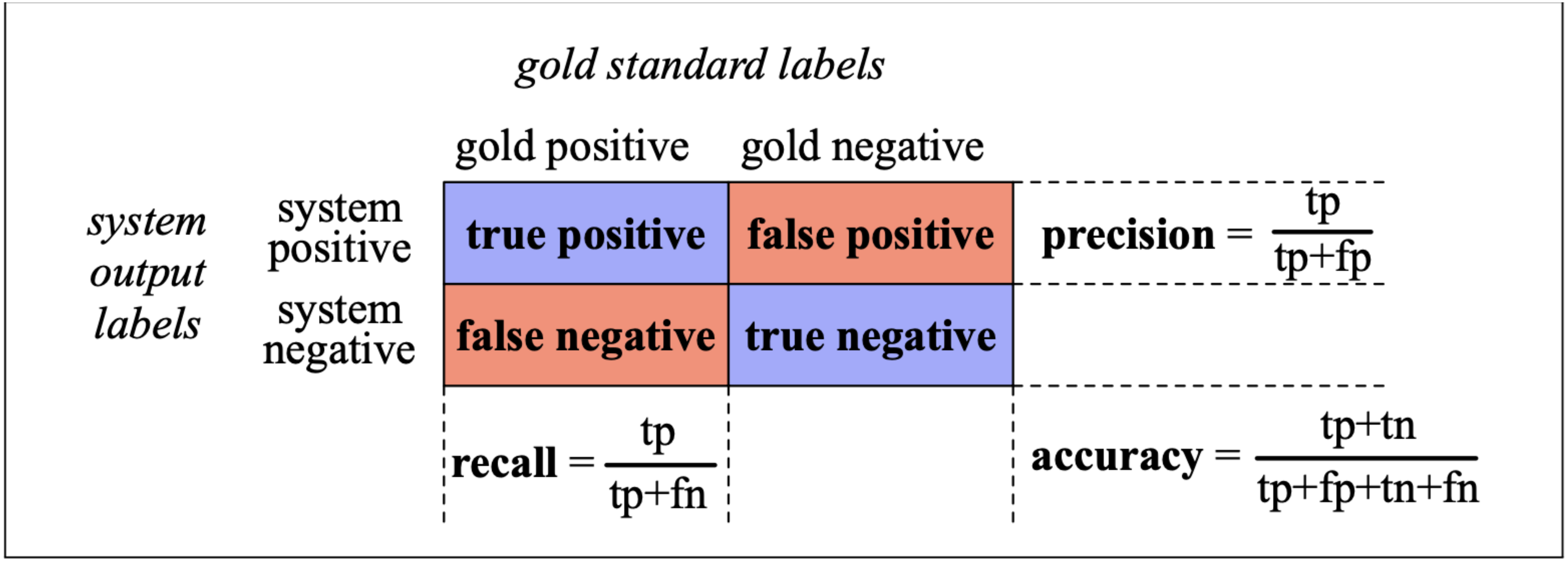


Figure 4.4 A confusion matrix for visualizing how well a binary classification system performs against gold standard labels.

Accuracy

- **Definition**: Accuracy is the ratio of correctly classified examples (both positives and negatives) to the total number of examples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Works well when**:
  - The dataset is **balanced** (equal number of positive and negative examples).
  - Both **positive and negative** cases are equally important.

- In many practical applications, **missing a positive case** (False Negative) can be much more costly than making a **false positive**.
- **Accuracy may be misleading** when the dataset is imbalanced (i.e., one class is much larger than the other).

Why accuracy not enough?

- **Scenario**: A medical test for a rare disease.
- **True Positives (TP)**: The test correctly identifies someone with the disease.
- **True Negatives (TN)**: The test correctly identifies someone without the disease.
- **False Positives (FP)**: The test incorrectly identifies someone without the disease as having it.
- **False Negatives (FN)**: The test fails to identify someone who actually has the disease.

- **Problem**: In disease detection, it is **much more important** to **catch all the people with the disease** (minimizing False Negatives) than to avoid falsely diagnosing someone as sick (False Positive).

  - **Example**: If the test misses a cancer case (False Negative), the person could go untreated, which is much more harmful than falsely diagnosing a healthy person as having cancer.

Accuracy vs Precision/Recall

- **Accuracy Calculation**:
  - If 95 out of 100 healthy people are correctly identified as healthy and 5 patients with cancer are missed (False Negatives), the accuracy could still be high (e.g., 95% accuracy).
- However, **missing cancer patients (FN)** can be much worse than falsely diagnosing a healthy person, which is why **precision and recall** are more meaningful in this context than accuracy alone.

Why Precision and Recall Matter More ?

- **Precision**: Focuses on reducing False Positives (FP), i.e., how many predicted positives are actually correct.
- **Recall**: Focuses on reducing False Negatives (FN), i.e., how many actual positives are identified.
- **In the disease detection example**, **high recall** (catching more positive cases) is more important than **high precision** (limiting false positives).

Precision

Precision

- **Definition**: Precision measures the percentage of items that the system labeled as **positive** that are actually **positive** according to the human **gold labels**.
- **Formula**:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

- **Explanation**:
  - High precision means that when the system classifies an item as positive, it is very likely to be correct (few False Positives).

Precision

- This metric measures the number of words in the Predicted Sentence that also occur in the Target Sentence.
- **Target Sentence**: He eats an apple
- **Predicted Sentence**: He ate an apple
- We would normally compute the Precision using the formula:
    - *Precision = Number of correct predicted words / Number of total predicted words*
    - *Precision = 3 / 4*
- But using Precision like this is not good enough. There are two cases that we still need to handle.

- **Repetition**
- The first issue is that this formula allows us to cheat. We could predict a sentence:
  - **Target Sentence**: He eats an apple
  - **Predicted Sentence**: He He He
- and get a perfect Precision = 3 / 3 = 1

- **Multiple Target Sentences**
- Secondly, as we’ve already discussed, there are many correct ways to express the same sentence. In many NLP models, we might be given multiple acceptable target sentences that capture these different variations.
- We account for these two scenarios using a modified Precision formula which we’ll call “Clipped Precision”.
- **Clipped Precision**
- Let’s go through an example to understand how it works.
- Let’s say, that we have the following sentences:
- **Target Sentence 1**: He eats a sweet apple
- **Target Sentence 2**: He is eating a tasty apple
- **Predicted Sentence**: He He He eats tasty fruit

We now do two things differently:

- We compare each word from the predicted sentence with all of the target sentences. If the word matches any target sentence, it is considered to be correct.
- We limit the count for each correct word to the maximum number of times that that word occurs in the Target Sentence. This helps to avoid the Repetition problem. This will become clearer below.

Word	***Matching Sentence***	***Matched Predicted Count***	***Clipped Count***
He	*Both*	*3*	*1*
eats	*Target 1*	*1*	*1*
tasty	*Target 2*	*1*	*1*
fruit	*None*	*0*	*0*
Total		***5***	***3***

Recall (or Sensitivity)

Recall

- **Definition**: Recall measures the percentage of items that are **actually positive** in the input and were correctly identified by the system.
- **Formula**:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

- **Explanation**:
    - High recall means that the system is good at identifying most of the actual positive cases (few False Negatives).

- **Precision**: Focuses on reducing False Positives (FP), i.e., how many predicted positives are actually correct.
  - **Precision** focuses on the accuracy of the **predicted positives**.

- **Recall**: Focuses on reducing False Negatives (FN), i.e., how many actual positives are identified.
  - **Recall** focuses on how many **actual positives** were detected.

Precision vs Recall Trade-off

- **High Precision** → Lower **False Positives** (fewer incorrectly labeled positive cases).
- **High Recall** → Lower **False Negatives** (fewer missed positive cases).
- **Trade-off**: Often, improving one will lead to a reduction in the other. The goal is to find a balance depending on the specific application.

When to Use Each

- **Precision**: Prioritize when you want to avoid False Positives (e.g., spam email detection, fraud detection).
- **Recall**: Prioritize when you want to minimize False Negatives (e.g., disease detection, identifying rare events).

Low Precision, High Recall

- The model is very good at identifying most of the actual positives but also makes a lot of false positive predictions.
- The model is too lenient, identifying almost all positive cases but with many false positives.
- **Interpretation**:
  - **Low precision** because many predicted positives are actually false positives.
  - **High recall** because most of the actual positive cases are identified.

Example case

- **Medical Test for a Rare Disease**:
  - The test is highly **sensitive** (high recall) and identifies almost everyone with the disease (true positives).
  - However, it might incorrectly label some healthy people as having the disease (false positives).
- **Spam Filter Example**:
  - It marks almost all spam emails as spam (high recall), but incorrectly flags legitimate emails as spam (low precision).

High Precision, Low Recall

- The model is conservative in predicting positives. It ensures that when it predicts a positive, it’s likely to be correct, but it misses many actual positives (false negatives).
- It prioritizes accuracy over quantity.

- **Interpretation**:
- **High precision** because almost all predicted positives are true positives.
- **Low recall** because many actual positives are missed.

Example case

- **Fraudulent Transactions**:
  - A model flags only transactions that are highly likely to be fraudulent (high precision, few false positives).
  - However, it misses many fraudulent transactions (low recall).

- **Cancer Detection**:
  - A medical test only classifies individuals as having cancer when it’s very confident, ensuring high precision but missing many actual cancer cases (low recall).

F-Score

F-measure (van Rijsbergen, 1975)

- **Definition**: The F-measure is a single metric that combines **precision** and **recall**. It is the **weighted harmonic mean** of precision and recall, giving a balance between the two metrics.
- **General Formula**:

$$F_\beta = \frac{(\beta^2 + 1) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

  - **β parameter**: Differentiates the importance of recall vs. precision:
    - **β > 1**: Favors **recall**.
    - **β < 1**: Favors **precision**.
    - **β = 1**: **Precision and recall are equally weighted**, known as **F1 score**.

F1 Score (β = 1)

- **Formula**:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

- **Explanation**: The **F1 score** is the most commonly used form of the F-measure, where precision and recall are equally important.

Weighted F-measure

- **Alternative Formula**:

$$F = \frac{(\beta^2 + 1) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

  - **β = 1 - α**: Allows further control over the weighting of precision and recall.

F1 Score (β = 1)

- **When to use**: **F1 score** is used when **precision** and **recall** are equally important.
- **Example Use Cases**:
  - **Information Retrieval** (Search Engines):
    - We want to retrieve the most relevant documents (high recall) but also want to ensure that retrieved documents are actually relevant (high precision).
    - A balance is necessary.
  - **Named Entity Recognition (NER)**:
    - In NER, the goal is to identify named entities (e.g., person names, organizations) correctly.
    - A balance of precision and recall is typically desired to ensure that both precision and recall are equally weighted.
  - **Sentiment Analysis**:
    - Identifying positive or negative sentiment in reviews or social media comments, where both correctly identifying sentiment and minimizing misclassification are important.

F2 Score (β = 2)

- **When to use**: **F2 score** gives more weight to **recall** than precision. This is useful when you want to **capture all positive cases**, even at the cost of some false positives.
  - **Example Use Cases**:
    - **Medical Text Classification**: When identifying medical conditions or diseases, it's often more important to catch **all** potential positive cases (e.g., patients with a rare disease), even if some healthy individuals are mistakenly labeled as sick. Here, recall is prioritized.
    - **Spam Detection**: In spam filters, it might be more important to **catch all spam emails** (high recall), even if some legitimate emails are marked as spam (low precision). False negatives (missing spam) are more harmful than false positives.

F0.5 Score (β = 0.5)

- **When to use**: **F0.5 score** gives more weight to **precision** than recall. This is useful when it's important to ensure that positive predictions are **correct** and we can afford missing some positives.
  - **Example Use Cases**:
    - **Legal Document Search**: In legal or technical document searches, it's more important to retrieve **highly relevant documents** (precision) rather than retrieving a larger number of documents, which may contain irrelevant information (low recall). Here, false positives (irrelevant documents) are more problematic than missing relevant ones.
    - **Fraud Detection**: In fraud detection, we might prefer **high precision** to ensure that the flagged transactions are actually fraudulent, even if some fraudulent transactions are missed (low recall). This prevents wasting resources on false alerts.

Evaluating with more than two classes

- What happens when more than two classes?
- For sentiment analysis we generally have 3 classes (positive, negative, neutral) and even more classes are common for tasks like part-of-speech tagging, word sense disambiguation, semantic role labeling, emotion detection, and so on.

		gold labels			
		urgent	normal	spam	
system output	urgent	8	10	1	**precision**$_u = \frac{8}{8+10+1}$
	normal	5	60	50	**precision**$_n = \frac{60}{5+60+50}$
	spam	3	30	200	**precision**$_s = \frac{200}{3+30+200}$
		recall$_u = \frac{8}{8+5+3}$	**recall**$_n = \frac{60}{10+60+30}$	**recall**$_s = \frac{200}{1+50+200}$	

Figure 4.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2.

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

$$\text{precision} = \frac{8}{8+11} = .42$$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

$$\text{precision} = \frac{60}{60+55} = .52$$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

$$\text{precision} = \frac{200}{200+33} = .86$$

Pooled

	true yes	true no
system yes	268	99
system no	99	635

$$\begin{array}{c}\text{microaverage}\\ \text{precision}\end{array} = \frac{268}{268+99} = \mathbf{.73}$$

$$\begin{array}{c}\text{macroaverage}\\ \text{precision}\end{array} = \frac{.42+.52+.86}{3} = \mathbf{.60}$$

Figure 4.6 Separate confusion matrices for the 3 classes from the previous figure, showing the pooled confusion matrix and the microaveraged and macroaveraged precision.

What is Microaverage Precision?

- **Microaverage precision** calculates precision globally by counting the total true positives, false positives, and false negatives across all classes.
- **Formula**:

$$\text{Microaverage Precision} = \frac{\sum TP}{\sum TP + \sum FP}$$

- **How it Works**:
  - **True Positives (TP)**: The sum of correctly predicted positives across all classes.
  - **False Positives (FP)**: The sum of incorrectly predicted positives across all classes.
- **Why Use Microaverage?**:
  - **Class Imbalance**: Microaverage is useful when the class distribution is imbalanced because it focuses on overall correctness. It's especially helpful when each individual class's performance is not as important, but rather the overall performance across all classes matters.

What is Macroaverage Precision?

- **Macroaverage precision** calculates precision for each class independently and then averages them. It gives equal weight to all classes, regardless of their frequency.
- **Formula**:

$$\text{Macroaverage Precision} = \frac{1}{N}\sum_{i=1}^{N}\frac{TP_i}{TP_i + FP_i}$$

  Where N is the number of classes.

- **How it Works**:
  - **TP_i** and **FP_i** are the true positives and false positives for the i^{th} class.
  - Precision is computed for each class, and then the average is taken.
- **Why Use Macroaverage?**:
  - **Equal Weight to All Classes**: Macroaverage is more useful when you care equally about the performance of each class, regardless of how frequently each class appears in the dataset.
  - **No Class Dominance**: It avoids the dominance of larger classes in the evaluation.

- Why we cannot use Precision and Recall for other tasks?

1. **Issues with Direct Word Matches**

- **Precision and recall** are based on **exact word matches**, which makes them less suitable for MT, where translations often involve **paraphrases**, **word reordering**, or **structural differences** between source and target languages.
  - For example, the sentence **"I like apples"** might be translated as **"Apples are liked by me"**, which is a **correct translation** but differs in word order.
- **Challenge**: A direct word-to-word match (for precision/recall) does not capture the **semantic meaning** or **syntactic structure** of the translated sentence.

2. Sentence-Level Evaluation

- **Precision** and **Recall** are often **calculated at the word level**.
  - In MT, **evaluation is better done at the sentence level**, where even small variations or word substitutions could affect the meaning.
  - For example:
    - **Source**: "She went to the market."
    - **Translation**: "She went shopping."
    - Even though the words are different, the meaning is similar.
- **Precision and recall** alone do not account for the **quality of translation** in terms of fluency, adequacy, or meaning preservation.

3. Lack of Context

- **Precision and recall** do not consider **context** or **grammar**. In MT, the context of the entire sentence is crucial to produce a fluent and accurate translation.
  - Example:
    - **Source**: "The bank is near the river."
    - **MT Output**: "The financial institution is close to the river."
    - While the words are different, the meaning and context are correctly conveyed, something precision/recall can't fully capture.